Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 1

Title: Importing business datasets from flat files & MySQL and connecting with any other sources into Power BI

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GitHub Link: https://github.com/abhi7abhi/BA

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Experiment No.: 1

Power BI Sales Performance Analysis – Sample Superstore

Problem Statement

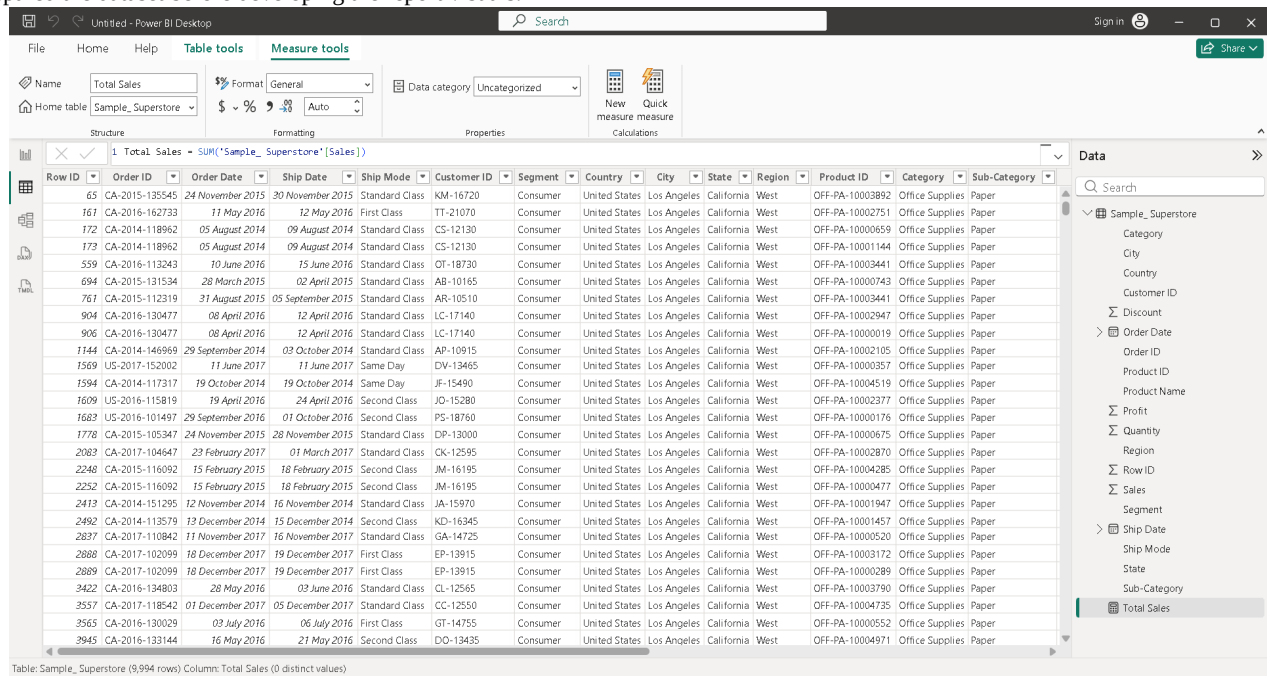
Management wants to quickly understand sales performance across products, categories, sub-categories, and regions. The objective is to prepare the Sample Superstore dataset in Power BI and build an interactive dashboard that makes the major business patterns easy to identify.

Dataset and Model Used

- The PBIX contains a single main table named Sample_ Superstore.
- The table includes Order Date, Ship Date, Ship Mode, Customer ID, Segment, Country, City, State, Region, Product ID, Category, Sub-Category, Product Name, Profit, Quantity and Sales.
- The PBIX includes the DAX measures Total Sales, Total Profit and Total Quantity.
- Analysis is performed directly from the Sample_ Superstore table.

Step 1: Data Preparation and Import

- Opened the Sample Superstore data in Power BI Desktop.
- Inspected the imported table in Data view.
- Verified the required numeric, categorical, product, region and date fields.
- Prepared the dataset before developing the report visuals.



Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Segment	Country	City	State	Region	Product ID	Category	Sub-Category
65	CA-2015-135545	24 November 2015	30 November 2015	Standard Class	KM-16720	Consumer	United States	Los Angeles	California	West	OFF-PA-10003892	Office Supplies	Paper
161	CA-2016-162733	11 May 2016	12 May 2016	First Class	TT-21070	Consumer	United States	Los Angeles	California	West	OFF-PA-10002751	Office Supplies	Paper
172	CA-2014-118962	05 August 2014	09 August 2014	Standard Class	CS-12130	Consumer	United States	Los Angeles	California	West	OFF-PA-10000659	Office Supplies	Paper
173	CA-2014-118962	05 August 2014	09 August 2014	Standard Class	CS-12130	Consumer	United States	Los Angeles	California	West	OFF-PA-10001144	Office Supplies	Paper
559	CA-2016-113243	10 June 2016	15 June 2016	Standard Class	OT-18730	Consumer	United States	Los Angeles	California	West	OFF-PA-10003441	Office Supplies	Paper
694	CA-2015-131534	28 March 2015	02 April 2015	Standard Class	AB-10165	Consumer	United States	Los Angeles	California	West	OFF-PA-10000743	Office Supplies	Paper
761	CA-2015-112319	31 August 2015	05 September 2015	Standard Class	AR-10510	Consumer	United States	Los Angeles	California	West	OFF-PA-10003441	Office Supplies	Paper
904	CA-2016-130477	08 April 2016	12 April 2016	Standard Class	LC-17140	Consumer	United States	Los Angeles	California	West	OFF-PA-10002947	Office Supplies	Paper
906	CA-2016-130477	08 April 2016	12 April 2016	Standard Class	LC-17140	Consumer	United States	Los Angeles	California	West	OFF-PA-10000019	Office Supplies	Paper
1144	CA-2014-146969	29 September 2014	03 October 2014	Standard Class	AP-10915	Consumer	United States	Los Angeles	California	West	OFF-PA-10002105	Office Supplies	Paper
1569	US-2017-152002	11 June 2017	11 June 2017	Same Day	DV-13465	Consumer	United States	Los Angeles	California	West	OFF-PA-10000357	Office Supplies	Paper
1594	CA-2014-117317	19 October 2014	19 October 2014	Same Day	JF-15490	Consumer	United States	Los Angeles	California	West	OFF-PA-10004519	Office Supplies	Paper
1609	US-2016-115819	19 April 2016	24 April 2016	Second Class	JO-15280	Consumer	United States	Los Angeles	California	West	OFF-PA-10002377	Office Supplies	Paper
1683	US-2016-101497	29 September 2016	01 October 2016	Second Class	PS-18760	Consumer	United States	Los Angeles	California	West	OFF-PA-10000176	Office Supplies	Paper
1778	CA-2015-105347	24 November 2015	28 November 2015	Standard Class	DP-13000	Consumer	United States	Los Angeles	California	West	OFF-PA-10000675	Office Supplies	Paper
2083	CA-2017-104647	23 February 2017	01 March 2017	Standard Class	CK-12595	Consumer	United States	Los Angeles	California	West	OFF-PA-10002870	Office Supplies	Paper
2248	CA-2015-116092	15 February 2015	18 February 2015	Second Class	JM-16195	Consumer	United States	Los Angeles	California	West	OFF-PA-10004285	Office Supplies	Paper
2252	CA-2015-116092	15 February 2015	18 February 2015	Second Class	JM-16195	Consumer	United States	Los Angeles	California	West	OFF-PA-10000477	Office Supplies	Paper
2413	CA-2014-151295	12 November 2014	16 November 2014	Standard Class	JA-15970	Consumer	United States	Los Angeles	California	West	OFF-PA-10001947	Office Supplies	Paper
2492	CA-2014-113579	13 December 2014	15 December 2014	Second Class	KD-16345	Consumer	United States	Los Angeles	California	West	OFF-PA-10001457	Office Supplies	Paper
2837	CA-2017-110842	11 November 2017	16 November 2017	Standard Class	GA-14725	Consumer	United States	Los Angeles	California	West	OFF-PA-10000520	Office Supplies	Paper
2888	CA-2017-102099	18 December 2017	19 December 2017	First Class	EP-13915	Consumer	United States	Los Angeles	California	West	OFF-PA-10003172	Office Supplies	Paper
2889	CA-2017-102099	18 December 2017	19 December 2017	First Class	EP-13915	Consumer	United States	Los Angeles	California	West	OFF-PA-10000289	Office Supplies	Paper
3422	CA-2016-134803	28 May 2016	03 June 2016	Standard Class	CL-12565	Consumer	United States	Los Angeles	California	West	OFF-PA-10003790	Office Supplies	Paper
3557	CA-2017-118542	01 December 2017	05 December 2017	Standard Class	CC-12550	Consumer	United States	Los Angeles	California	West	OFF-PA-10004735	Office Supplies	Paper
3565	CA-2016-130029	03 July 2016	06 July 2016	First Class	GT-14755	Consumer	United States	Los Angeles	California	West	OFF-PA-10000552	Office Supplies	Paper
3945	CA-2016-133144	16 May 2016	21 May 2016	Second Class	DO-13435	Consumer	United States	Los Angeles	California	West	OFF-PA-10004971	Office Supplies	Paper

Figure 1: Sample_ Superstore table and Total Sales measure

Step 2: Data Cleaning and Transformation

- Opened Power Query Editor to inspect the Sample_ Superstore query.
- Checked the initial column types after headers were promoted.
- Converted date fields to appropriate Date types for analysis.
- The Ship Date transformation uses the English (United States) locale.
- Reviewed the transformed data preview before loading it into the report.

The Power Query screenshots show the preparation stage. Order Date and Ship Date are treated as dates, while Order ID, Ship Mode, Customer ID and Segment remain categorical/text fields.

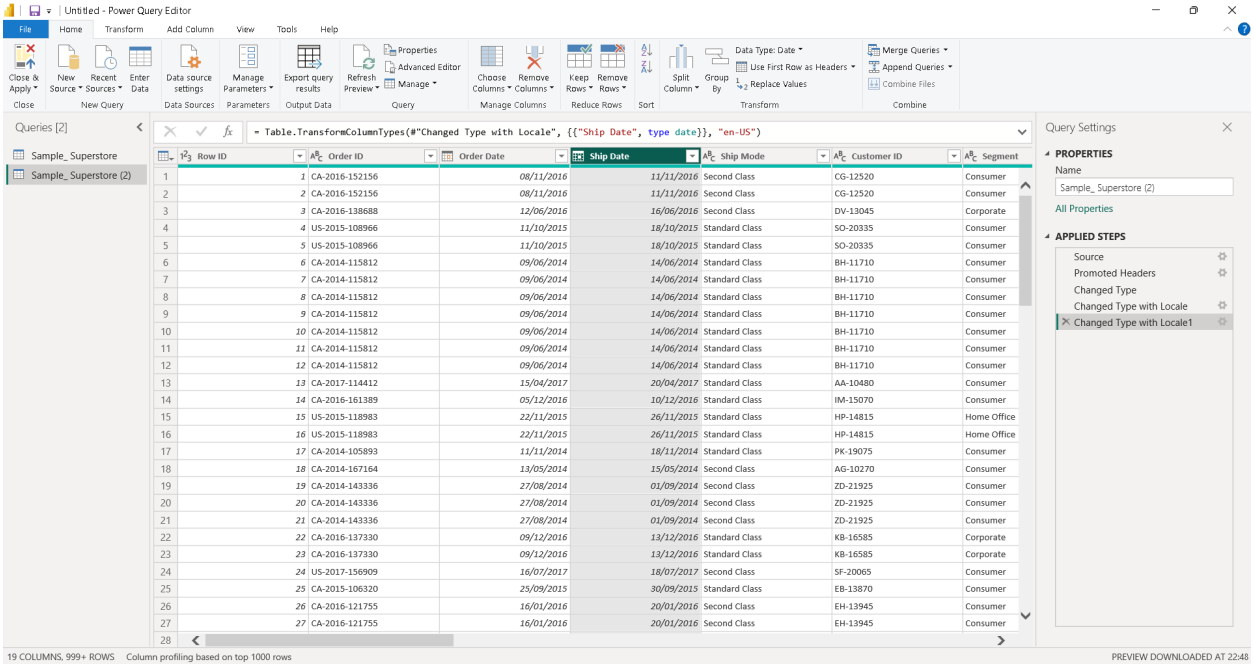


Figure 2: Power Query – Ship Date date-type transformation

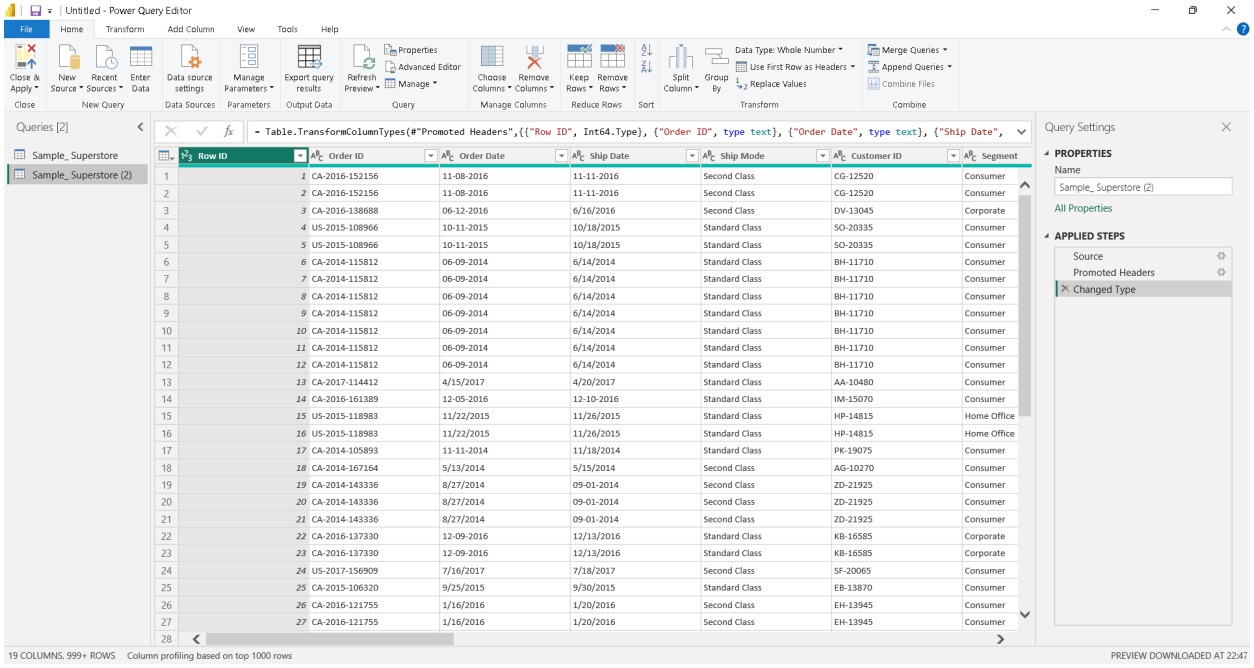


Figure 3: Power Query – promoted headers and changed types

Step 3: DAX Measures

- Total Sales – sum of Sales.
- Total Profit – sum of Profit.
- Total Quantity – sum of Quantity.

Step 4: Power BI Visualizations

The dashboard was developed to answer management questions about overall performance, category performance, regional performance, products and time trends.

- Total Sales by Category – horizontal bar chart.
- Total Sales by Region – column chart.
- Total Sales by Sub-Category – 100% stacked bar chart.
- Total Sales by Product Name – bar chart.
- Total Sales and Total Profit – scatter chart.
- Total Sales by Year – line chart.
- Region, Year and Category – interactive slicers.
- KPI cards – Total Sales, Total Profit and Total Quantity.

Visualization Pictures

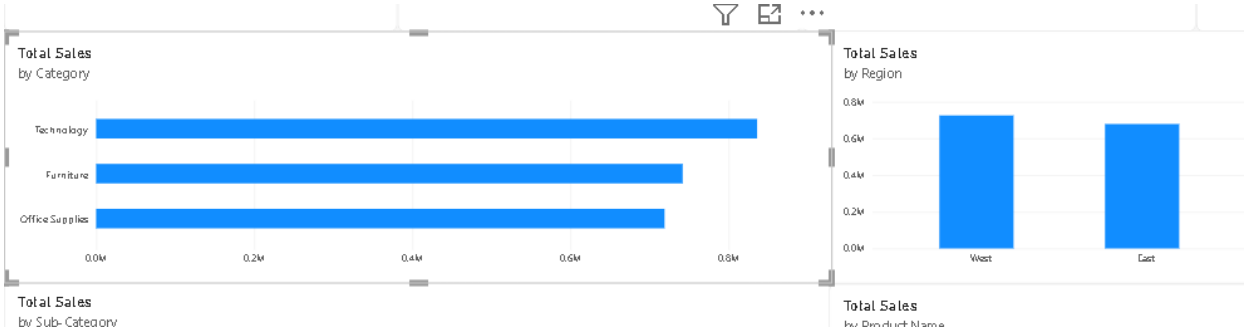


Figure 4: Sales by Category and Sales by Region

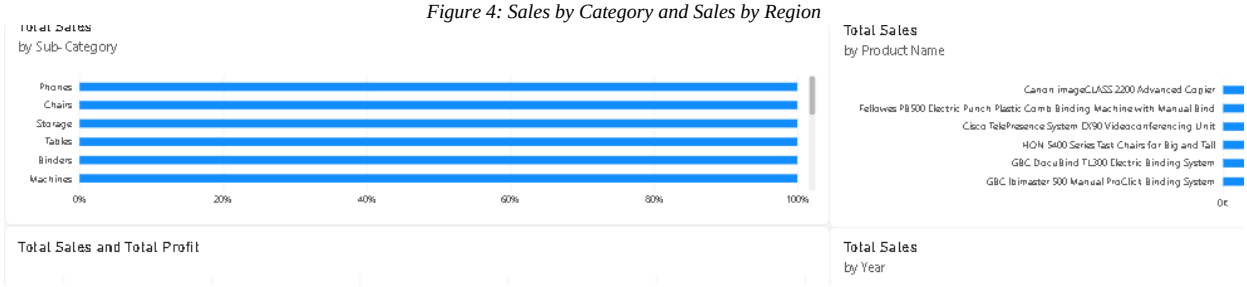


Figure 5: Sales by Sub-Category and Product Name

Step 5: Dashboard Development and Output

The final PBIX page combines KPI cards, comparative charts and interactive slicers into one management dashboard.



Figure 6: Complete Sample Superstore Power BI dashboard

Dashboard KPI Values

- Total Sales: 2.30M
- Total Profit: 286.40K
- Total Quantity: 38K

Trend and Interactive Filtering

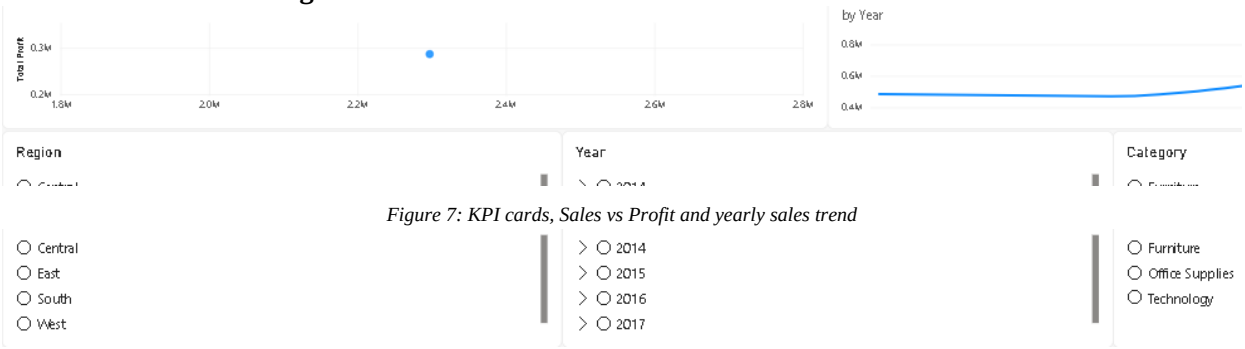


Figure 7: KPI cards, Sales vs Profit and yearly sales trend

Figure 8: Region, Year and Category slicers

Business Insights, Conclusion and Learning Outcomes

Observed Patterns

- Technology is the highest-selling category in the displayed category chart, followed by Furniture and Office Supplies.
- West has the highest regional sales, followed by East, Central and South.
- The Product Name chart identifies leading products by Total Sales.
- The yearly line chart shows changes in Total Sales over time.
- The scatter visual compares Total Sales with Total Profit.

Business Use

- Management can compare category performance quickly.
- Regional performance can be evaluated using the Region visual and slicer.
- Product-level sales can be used to identify leading products.
- Year and Category slicers allow focused analysis.

Conclusion

This project successfully uses the Sample Superstore dataset to create an interactive sales-performance report in Power BI. Data preparation was performed in Power Query, DAX measures were used for key metrics, and multiple visualization types were selected to answer business questions. The final dashboard combines sales, profit, quantity, category, sub-category, product, regional and yearly analysis.

Learning Outcomes

- Learned Power BI data inspection and preparation.
- Learned Power Query data-type and transformation tasks.
- Learned to create and use DAX measures.
- Learned to create KPI cards, charts and slicers.
- Learned to interpret sales patterns across categories, products, regions and years.

Final Result

The completed PBIX report provides management with a centralized and interactive view of Sample Superstore sales performance and supports quick comparison across products, categories, sub-categories, regions and years.